

dr. kornpob bhirombhakdi

AI Security Researcher, Educator, and Consultant — dual PhDs (Astrophysics + Mechanism Design & Game Theory), NASA postdoc, red-teaming lead, author of the *Spellbook of Jailbreak and Agent Vulnerabilities*. High-stakes AI safety, agent vulnerabilities, adversarial ML.

- dual PhDs + NASA postdoc
- red-teaming / adversarial research lead
- author, *Spellbook of Jailbreak and Agent Vulnerabilities*
- Winner, *The Judgement Day 2026 Phase-1 Scenario Submission*

Quick Links

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recent bits

- **2600TH Red x Blue Pill 2026 (talk): jailbreaks, agent vulnerabilities, and the grey mage tower** — [github](#)
- **Spellbook of Jailbreak and Agent Vulnerabilities** — [Zenodo:21665118](#) · [Page](#) · [Spotify](#) · [Youtube](#)
- **The Judgement Day 2026 Phase-1 Scenario Submission** — [GitHub](#) · [Spotify](#) · [Youtube](#)
- **Goodboy Recon** — [Github](#) · [Medium](#)
- **ZADDY breadcrumbs 2026 collection** — [Zenodo:21791515](#) · [Spotify](#) · [Substack](#)

Education

- **Ph.D. Physics** | Ohio University, Athens, OH USA | 2014-2019
Light Curve Powering Mechanism of Superluminous Supernovae
- **Ph.D. Economics** | Chulalongkorn University, Bangkok TH | 2009-2013
Game Theory & Experimental Studies on Reciprocity
- **M.S. Physics** | Ohio University, Athens, OH USA | 2014-2015 | GPA: 3.951
- **Physics Pre-Degree** | Ramkhamhaeng University, Bangkok TH | 2013-2014 | GPA: 4.0
- **M.S. Health Economics** | Chulalongkorn University, Bangkok TH | 2007-2009
Technical Efficiency of University Hospitals
- **B.S. Medical Science** | Siriraj Hospital, Mahidol University, Bangkok TH | 2003-2007
Second Class Honours

Certifications and Training

- **Quantum Computing** | IBM | 2026 *Using a Quantum Computer Basics of Quantum Information*
- **Global MOOC on the Ethics of AI** | unesco | Sep 2026 | ID: 46ZWJV9V0OPK
- **Certified in Cybersecurity** | ISC2 | Jan 2026 - Dec 2028 | ID: 3153050
- **BASI Research Contributor** | Oct 2025 - present
- **Cybersecurity Trainings** | THNCA+NCSA Thailand 2025 *CompTIA Cloud+, Computer Hacking Forensic Investigator, Basic Penetration Testing*

- **Cybersecurity Professional Course** | NCSA Thailand | 2025 | ID: 5241057415KP
- **Basic Cybersecurity** | NCSA Thailand | 2025 | ID: SPG-ZNEXJM1UDTRGX
- **Deep Learning Specialization** | DeepLearning.AI (Coursera) | 2025 | ID: GYIBDXJ9UMCP
- **Data Science and Machine Learning Specialization** | Johns Hopkins University (Coursera) | 2015 – 2017 Courses: *Developing Data Products, Practical Machine Learning, Regression Models, Exploratory Data Analysis, Getting and Cleaning Data, Reproducible Research, R Programming, Statistical Inference, The Data Scientist's Toolbox*

Employment

- **On-Demand Consultant — Business Analytics and AI Solutions** | DR.KB (Self-Employed), Bangkok, Thailand | Sep 2025 – Present

Consult C-suite and product leads on AI integration, risk mitigation, and enterprise analytics roadmaps. Design, build, and deploy custom machine learning models, LLM agents, and automated data pipelines. Bridge technical engineering and executive decision-making to drive ROI and project execution.
- **Lead Prompt Security Researcher** | InjectPrompt.com (Contract), Remote | Jan 2026 – May 2026
Led red-team research on AI agent safety and jailbreak tactics. Designed custom RAG knowledge bases, skills, and orchestration workflows for sandboxed jailbreak testing. Authored and released *InjectPrompt Companion 3.0* and the *Jailbreak Perturbation Framework (JPF)*.
- **Data Scientist** | Sertis Co., Ltd., Bangkok, Thailand | Aug 2024 – Aug 2025
Developed deep learning architectures for computer vision and multi-object tracking. Built scalable multi-branch forecasting models for retail and wholesale supply chains. Maintained and optimized enterprise-grade ML assets. Delivered expert consulting and pitched high-value extension solutions.
- **Postdoctoral Researcher** | Space Telescope Science Institute (AURA/NASA), Baltimore, MD, USA | Aug 2019 – Feb 2024
- **Research/Teaching Assistant** | Physics and Astronomy Department, Ohio University, Athens, OH, USA | Aug 2014 – May 2019
- **Senior Consultant** | LEARN Corporation Co., Ltd., Bangkok, Thailand | Jan 2012 – Jan 2014
- **Market Research Consultant** | Freelance, Bangkok, Thailand | Jan 2010 – Jan 2012
- **Coordinator Assistant** | Economic Impacts of Demographic Change Project, Thailand Research Fund, Bangkok, Thailand | Jan 2011 – Jan 2013
- **Lecturer** | 2010, 2012 | Faculty of Hospitality and Tourism, Prince of Songkla University, Phuket, Thailand
 - Principles of Economics (Undergraduate Level; International Program)
- **Lecturer** | 2009 | Faculty of Economics, Khonkaen University, Nongkhai, Thailand
 - Health Economics (Undergraduate Level)
- **Mathematician** | Institute of Academic Development Inc., Bangkok, Thailand | 2009
- **Teaching Assistant** | Faculty of Economics, Chulalongkorn University, Bangkok, Thailand
 - Advanced Microeconomic Theory II (Ph.D. Level; International Program) — 2010, 2011
 - Statistics for Health Economics (Graduate Level; International Program) — 2011
 - Introduction to Econometrics (Undergraduate Level) — 2010 – 2012
 - Introduction to Economics (Undergraduate Level) — 2010
 - Math for Economist I (Undergraduate Level) — 2009

- Math for Economist II (Undergraduate Level) — 2012

- **Consultant for Business Research and Development** | Freelance, Bangkok, Thailand | 2007 – 2014
- **Tutor** | Freelance, Bangkok, Thailand | 2007 – 2014
- **Actor/Model/DJ/Brand Ambassador** | Freelance, Bangkok, Thailand | 2005 – 2010 | IMDb

Research and Publications

Software and Applications

- **InjectPrompt Companion 3.0** | companion.injectprompt.com | Structured prompt engineering and systematic extraction methodologies for safety testing. Multi-tactic jailbreak templates following the Jailbreak Perturbation Framework (JPF). A progressive learning curriculum powering research from beginner through advanced levels.
- **hstgrism** | pypi.org/project/hstgrism/ | Python 3 package for reducing HST grism observations, providing a modular, aXe-compatible workflow with simpler installation, diagnostics, and flexible point-source extraction. Optimized for flexible point-source extraction and 2D polynomial background modeling. Powering data reduction pipelines in peer-reviewed observational studies, including Bhirombhakdi et al. (2024).
- **EchoEcho** | github.com/OhioUniversityGameDevelopersAssociation/EchoEcho | Help Fernando the Bat echolocate the way out and survive attacks from mad stone gargoyles. A prototype from Global Game Jam 2017, together with the Ohio University Game Developers Association (OUGDA), exploring wave and darkness.

Zines and Audios

- **ZADDY breadcrumbs 2026 collection** | Zenodo:21791515 · Page · Spotify · Youtube
- **spellbook of jailbreak and agent vulnerabilities** | Zenodo:21665118 · Page · Spotify · Youtube
- **spellbook of task avoidance wards** | Zenodo:20821697 · Page · Spotify · Youtube

Online Hubs

- linkedin: linkedin.com/in/bkornpob
- multiverselib-collectives: bkornpob.github.io
- old-dr-kb-hub: kbhirombhakdi.weebly.com
- substack: substack.com/@bkornpob
- medium: dr-kb.medium.com

Publications and Talks

- **Talk: jailbreaks, agent vulnerabilities, and the grey mage tower** | (2026) 2600TH Red x Blue Pill 2026, Bangkok, Thailand
- **ZADDY Breadcrumbs 2026 Collection** | Bhirombhakdi, K. (2026) zenodo:21791515. [Link](#)
- **Spellbook of Jailbreak and Agent Vulnerabilities** | Bhirombhakdi, K. (2026) zenodo:21665118. [Link](#)
- **Spellbook of Task Avoidance Wards** | Bhirombhakdi, K. (2026) zenodo:20821697. [Link](#)
- **Seven Runes of Clarity: Hold Your Agency When Working with AI Agents** | Bhirombhakdi, K. (2026) zenodo:20638949. [Link](#)

- **Jailbreak Perturbation Framework: Theory and Applications** | Bhirombhakdi, K., Willis-Owen, D., and N, K. (2026) zenodo:20040972. [Link](#)
- **>dr.kb< 2025 LOOKBACK: Living Inside the Glitch** | Bhirombhakdi, K. (2026) zenodo:19825030. [Link](#)
- **An afterglow study of the “New Year’s Burst” GRB 220101A** | with Roychowdhury et al. (2026) arXiv:2602.04660. [Link](#)
- **Probing the distant universe through GRB afterglow spectroscopy** | with Tanvir et al. (2025) JWST Proposal, Cycle 4, ID. #8539. [Link](#)
- **The Redshift of GRB 190829A/SN 2019oyw: A Case Study of GRB-SN Evolution** | Bhirombhakdi et al. (2024) The Astrophysical Journal, Volume 977, Issue 2, id.256, 16 pp. [Link](#)
- **A Hubble Space Telescope Search for r-Process Nucleosynthesis in Gamma-ray Burst Supernovae** | with Rastinejad et al. (2024) The Astrophysical Journal, Volume 968, Issue 1, id.14, 17 pp. [Link](#)
- **Heavy-element Production in a Compact Object Merger Observed by JWST** | with Levan et al. (2024) Nature, Volume 626, Issue 8000, p. 737–741. [Link](#)
- **A Long-duration Gamma-ray Burst of Dynamical Origin from the Nucleus of an Ancient Galaxy** | with Levan et al. (2023) Nature Astronomy, Volume 7, p. 976–985. [Link](#)
- **Panning for Gold, but Finding Helium: Discovery of the Ultra-stripped Supernova SN2019wxt from Gravitational-wave Follow-up Observations** | with ENGRAVE Collaboration (2023) Astronomy and Astrophysics, Volume 675, id.A201, 34 pp. [Link](#)
- **The First JWST Spectrum of a GRB Afterglow: No Bright Supernova in Observations of the Brightest GRB of all Time, GRB 221009A** | with Levan et al. (2023) The Astrophysical Journal Letters, Volume 946, Issue 1, id.L28, 14 pp. [Link](#)
- **An Extremely Energetic Supernova from a Very Massive Star in a Dense Medium** | with Nicholl et al. (2020) Nature Astronomy, Volume 4, p. 893–899. [Link](#)
- **Observational Constraints on the Optical and Near-infrared Emission from the Neutron Star-Black Hole Binary Merger Candidate S190814bv** | with Ackley et al. (2020) Astronomy and Astrophysics, Volume 643, id.A113, 48 pp. [Link](#)
- **SN2008es: Strong Interacting Hydrogen-rich Superluminous Supernova Without Narrow H-alpha Features and Early Dust Formation** | Bhirombhakdi (2019) STScI Newsletters, Volume 36, Issue 03. [Link](#)
- **The Type II Superluminous SN2008es at Late Times: Near-infrared Excess and Circumstellar Interaction** | Bhirombhakdi et al. (2019) Monthly Notices of the Royal Astronomical Society, Volume 488, Issue 3, p. 3783–3793. [Link](#)
- **Light Curve Powering Mechanisms of Superluminous Supernovae** | Bhirombhakdi (2019) Dissertation, Ohio University, Athens, OH, USA. [Link](#)
- **Talk: Hunting Magnetar Central Engines in Superluminous Supernovae** | (2019) American Astronomical Society, AAS Meeting #233, id.410.04. [Link](#)
- **Where is the Engine Hiding its Missing Energy? Constraints from a Deep X-ray Non-detection of Superluminous SN2015bn** | Bhirombhakdi et al. (2018) The Astrophysical Journal Letters, Volume 868, Issue 2, article id. L32, 7 pp. [Link](#)
- **One Thousand Days of SN2015bn: HST Imaging Shows a Light Curve Flattening Consistent with Magnetar Predictions** | with Nicholl et al. (2018) The Astrophysical Journal Letters, Volume 866, Issue 2, article id. L24, 7 pp. [Link](#)

- **The Type I Superluminous Supernova PS16aqv: Lightcurve Complexity and Deep Limits on Radioactive Ejecta in a Fast Event** | with Blanchard et al. (2018) The Astrophysical Journal, Volume 865, Issue 1, article id. 9, 17 pp. [Link](#)
- **Spectroscopic Classification of Two Superluminous Supernovae** | with Chornock et al. (2016) The Astronomer's Telegram #8790. [Link](#)
- **Essays in Experimental Studies on Positive Reciprocity and Auction Design for an Object with Countervailing Positive Externalities** | Bhirombhakdi (2013) Dissertation, Chulalongkorn University, Bangkok, Thailand. [Link](#)
- **Cost of Action, Perceived Intention, Positive Reciprocity, and Signalling Model** | Bhirombhakdi and Potipiti (2012) Journal of Business and Policy Research, Volume 7, Number 3, September 2012 Special Issue, ISSN: 1449-387X. [Link to preprint version](#)
- **Talk: Cost of Action, Perceived Intention, Positive Reciprocity, and Signalling Model** | (2012) The 6th Asian Business Research Conference, Bangkok, Thailand
- **Performance of a Reciprocity Model in Predicting a Positive Reciprocity Decision** | Bhirombhakdi and Potipiti (2011) Chulalongkorn Journal of Economics, Vol. 23, No. 1. [Link](#)
- **Talk: Performance of a Reciprocity Model in Predicting a Positive Reciprocity Decision** | (2011) International Conference on Management, Economics and Social Sciences (ICMESS'2011), Bangkok, Thailand
- **Talk: Neuroeconomics** | (2010, 2011) Faculty of Economics, Chulalongkorn University, Bangkok, Thailand
- **Special Lecture: Opportunity Cost and Economic Evaluation** | (2010) The College of Public Health Science, Chulalongkorn University, Bangkok, Thailand
- **Special Lecture: Efficiency Analysis** | (2010) The College of Public Health Science, Chulalongkorn University, Bangkok, Thailand
- **Special Lecture: National Examination Preparation (Physics)** | (2009) (in Thai) organized by Buriram Provincial Administration Organization, Buriram, Thailand
- **Special Lecture: National Examination Preparation (Physics)** | (2009) Sarawittaya School, Bangkok, Thailand
- **Technical Efficiency of University Hospitals in Thailand** | Bhirombhakdi (2008) Thesis, Chulalongkorn University, Bangkok, Thailand. [Link](#)

Achievements

- **Winner, The Judgement Day — Call for Scenario Challenge (Track 1)** | 2026 Scenario: Counter-UAS System — Friendly Fire Decision (IFF Corruption Attack). [Link](#)
- **Space Telescope Proposals** | 2026 HST 14 proposals, JWST 6 proposals.
- **ENGRAVE - Electromagnetic counterparts of gravitational wave sources at the Very Large Telescope** International collaboration bringing together astronomers who use the ESO facilities to research gravitational wave events. [Link](#)
- **Bravo! Awards** | Space Telescope Science Institute (AURA/NASA) Hosting the Loyola University Physics Club on February 20, 2020. Contributing in Astrogrism Sprint during May 12-21, 2020.
- **Graduate Awards** | 2014 - 2019 Ohio University, Athens, OH, USA
- **Global Game Jam 2017** | 2017 Lead Programmer and Project Coordinator in the project "EchoEcho" with a team at GRID Lab, Ohio University, Athens, OH, USA. [Link](#)

- **Kendo 2-Dan** | 2017 All United States Kendo Federation, USA (AUSKF ID: 017666)
- **Executive Course on Social Security** | 2012 Representative from the Faculty of Economics, Chulalongkorn University, organized by the International Labour Organization (UN), in collaboration with Chulalongkorn University, Bangkok, Thailand
- **Hands-on Training “How to Cost, Finance and Monitor Social Protection Schemes?”** | 2012 Representative from the Faculty of Economics, Chulalongkorn University, organized by the International Labour Organization (UN), in collaboration with Chulalongkorn University and Health Insurance Research Institute of Thailand, Bangkok, Thailand
- **The Best Paper Awards** | 2012 “Cost of Action, Perceived Intention, Positive Reciprocity, and Signalling Model” at The 6th Asian Business Research Conference, 8–10 April 2012, Bangkok, Thailand.
- **Associate Fellowship** | 2012 World Academy of Social Science, Victoria, Australia
- **Research Grants** | 2012 The 90th Anniversary of Chulalongkorn University Fund (Ratchadaphiseksomphot Endowment Fund), Chulalongkorn University, Bangkok, Thailand
- **Brand Ambassador** | 2008 Playsmart Idol, organized by NC True Co., Ltd., Thailand. [Link](#)
- **Medical Scholarship Program** | 2004 – 2013 Mahidol University, Bangkok, Thailand
- **Gold Medal** | 2002 National Physics Olympiad Competition, The Promotion of Academic Olympiads, Thailand
- **High Distinction Awards** | 2001 Australian National Chemistry Quiz, senior division, Royal Australian Chemical Institute
- **Thailand Representative** | 2001 Arirang Youth Camp 5th, UNESCO, South Korea

Services

- **Independent AI Security and Red-Teaming Researcher and Contributor** | Oct 2025 – Present BASI
- **Member of Slitless Spectroscopy Group** | 2019 – 2024 Space Telescope Science Institute (AURA/NASA), Baltimore, MD, USA
- **Member of Review Editors** | 2023 Frontiers in Astronomy and Space Sciences. [Link](#)
- **Representative of Postdoctoral Fellows** | 2019 – 2023 Research Computing Forum at the Space Telescope Science Institute (AURA/NASA), Baltimore, MD, USA
- **Support Astronomer** | 2020 Seo Yoon Soyoona Kim in her work “Hearing Stars - SN1054 (Supernova)”. [Link](#)
- **Organizer** | 2020 – 2021 The Compact Objects and Supernovae Journal Club at the Space Telescope Science Institute (AURA/NASA), Baltimore, MD, USA
- **Leveler** | 2021 Hubble Space Telescope Cycle 29’s time allocation process. [Link](#)
- **Scientific Software Developer** | 2020 Kickoff sprint for Astrogrism at the Space Telescope Science Institute (AURA/NASA), Baltimore, MD, USA. [Link](#)
- **Member of the Host** | 2019 Visit by the Loyola University Physics Club at the Space Telescope Science Institute (AURA/NASA), Baltimore, MD, USA
- **Member of the Judges** | 2019 Chambliss Astronomy Achievement Student Awards at the American Astronomical Society, AAS Meeting #233
- **Member of the Judges** | 2018 Science Fair at the West Elementary, Athens, OH, USA

- **Proposal Reviewer Committee** | 2018 Original Work Grant and Travel Awards, in collaboration with The Graduate College and the Graduate Student Senate at Ohio University, Athens, OH, USA
- **Kendo Demonstration** | 2014 - 2018 Various community events, in collaboration with the Ohio University Kendo Club, Ohio University, Athens, OH, USA
- **LGBT Affairs Commissioner** | 2017 Graduate Student Senate, Ohio University, Athens, OH, USA
- **Science Staff** | 2015, 2017 Physics and Astronomy Open House organized by the Physics and Astronomy Department, Ohio University, Athens, OH, USA
- **Volunteer** | 2016 Athens Beautification Day, Athens, OH, USA
- **Science Staff** | 2014 Ohio Valley Museum of Discovery, Athens, OH, USA